SE444 Mid-Term II Grading Guide – Fall 2025
Total: 20 points (Part I: 4pts, Part II: 10pts, Part III: 6pts)

Part I: Multiple Choice (4 pts)

Partial credit for partially correct answers

Q1 (FOL, 1pt): A, C

- A: $\exists x\, Q(x)$ – From $\exists x\, P(x)$ and $\forall x\, (P(x) \to Q(x))$
- C: $\exists x\, (P(x) \land Q(x))$ – Same reasoning
- *NOT B*: Can't infer $\forall x\, Q(x)$
- *NOT D*: Converse not entailed
- *NOT E*: No info about $Q \land \neg P$

Q2 (Propositional, 1pt): C only

- From (3): $s = F, t = F$
- From (2): $r \to (s \lor t)$ forces $r = F$
- From (4): $q = T$
- From (1): $(p \land q) \to r$ with $q = T, r = F$ forces $p = F$
- Answer: KB $\models \neg p$

Q3 (CSP, 1pt): A, B, C, E

- Initial: $D(A) = \{1, 2\}, D(B) = \{1\}, D(C) = \{1, 2\}$
- A: TRUE – $B < C$ with $B = 1$ prunes C to $\{2\}$
- B: TRUE – $A \neq B$ with $B = 1$ prunes A to $\{2\}$
- C: TRUE – All domains become singletons
- D: FALSE – $B \leq C$ wouldn't prune C
- E: TRUE – AC-3 still prunes B to $\{1\}$ then C to $\{2\}$

Q4 (Agents, 1pt): B, D

- B: TRUE – Model-based maintains internal state for partial observability
- D: TRUE – Utility-based can value information-gathering actions
- *NOT A*: Simple reflex can't handle partial observability
- *NOT C*: Goal-based can handle stochastic environments
- *NOT E*: Table-driven infeasible for infinite/stochastic spaces

Part II: Logical Reasoning (10 pts)

Q1 (FOL, 3pts):

$$\forall s, t, c\, (\text{Student}(s) \land \text{Teacher}(t) \land \text{Course}(c) \land$$

$$\text{Takes}(s, c) \land \text{Teaches}(t, c)) \to$$

$$(\exists u\, (\text{Topic}(u) \land \text{Covers}(c, u) \land \text{Understands}(s, u)))$$

Grade: 3pts for correct, 2pts if minor quantifier issues, 1pt if structure partially correct

Q2 (FOL with Negation, 3pts): Accept any equivalent form:

$$\forall t, s\, ((\text{Teacher}(t) \land \text{Student}(s) \land \text{GradedBy}(s, t)) \to$$

$$(\exists c\, (\text{Course}(c) \land \text{Teaches}(t, c) \land \text{Takes}(s, c))))$$

Or contrapositive, negated-existence, or other equivalent forms

Q3 (CNF, 4pts):

1. Negate: $\neg\forall x\, (\text{Student}(x) \to \text{Hardworking}(x))$
2. Move $\neg$ in: $\exists x\, \neg(\text{Student}(x) \to \text{Hardworking}(x))$
3. Remove $\to$: $\exists x\, (\text{Student}(x) \land \neg\text{Hardworking}(x))$
4. Skolemize: $\text{Student}(a) \land \neg\text{Hardworking}(a)$

Grade: 1pt per step, must show transformations

Part III: CSP (6 pts)

Q1 (Formulation, 2pts):

- **Variables:** L_1, L_2, L_3 (0.5pt)
- **Domains:** $D(L_1) = \{A, C\}$, $D(L_2) = \{A, B, C\}$, $D(L_3) = \{B, C\}$ (1pt)
- **Constraints:** $L_1 \neq L_2$ (Labs 1&2 overlap), $L_2 \neq L_3$ (Labs 2&3 overlap) (0.5pt)

Q2 (AC-3, 3pts): Key Answer: NO PRUNING occurs

- All values in all domains have support (1.5pts)
- Must show checking at least 2 arcs (1pt)
- Final domains: $D(L_1) = \{A, C\}$, $D(L_2) = \{A, B, C\}$, $D(L_3) = \{B, C\}$ (0.5pt)
- *Partial: 0.5-1.5pts if incorrect pruning but logical reasoning shown*

Q3 (Solution, 1pt): Accept ANY valid assignment:

- $(L_1 = A, L_2 = B, L_3 = C)$ or $(L_1 = A, L_2 = C, L_3 = B)$
- $(L_1 = C, L_2 = A, L_3 = B)$ or $(L_1 = C, L_2 = B, L_3 = C)$
- Must satisfy: $L_1 \neq L_2$ and $L_2 \neq L_3$ and be in domains

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Common Mistakes to Watch:

- **Q1 (Part I):** Selecting D (converse fallacy) or B (overgeneralization)
- **Q2 (Part I):** Not tracing backward from constraint (3) through (2) to find $p = F$
- **Q2 (Part II):** Using $\neg$ GradedBy instead of implication structure
- **Q3 (Part II):** Missing Skolemization step or not showing all transformations
- **Q2 (Part III):** Incorrectly pruning values when AC-3 doesn't prune anything